Data Analysis & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean, median, mode and range

Find the mean of $6, 9, 12, 15, 18$.

Mean $= \frac{6+9+12+15+18}{5} = \frac{60}{5} = 12$.

2 [Foundational] • Target: Mean, median, mode and range

Ages of 7 students: $13, 15, 13, 16, 14, 13, 17$. Find the mean, median, and mode.

Ordered: $13, 13, 13, 14, 15, 16, 17$. Mean $= \frac{101}{7} \approx 14.43$. Median (4th value) $= 14$. Mode $= 13$ (appears three times).

3 [Foundational] • Target: Mean, median, mode and range

Find the range and mode of the data set $8, 3, 8, 15, 8, 22, 3$.

Range $= 22 - 3 = 19$. Mode $= 8$ (appears three times).

4 [Foundational] • Target: Mean, median, mode and range

A data set has mean 18 and 6 values. Find the sum of the values.

Sum $= 18 \times 6 = 108$.

5 [Foundational] • Target: Mean, median, mode and range

The data set $10, 12, 11, 13, 50$ contains an outlier. Find the mean before and after removing 50.

Before: $\frac{10+12+11+13+50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.

6 [Foundational] • Target: Mean, median, mode and range

A data set of 5 values has mean 20. Four of the values are $15, 18, 22, 25$. Find the fifth value.

Total sum $= 20 \times 5 = 100$. Known sum $= 15 + 18 + 22 + 25 = 80$. Fifth value $= 100 - 80 = 20$.

7 [Foundational] • Target: Reading tables

A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?

$\frac{12}{50} = \frac{6}{25}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Reading tables**

Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?

$15 + 13 = 28$ people. $\frac{28}{50} \times 100\% = 56\%$.

9 **[Intermediate]** • **Target: Reading tables**

A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.

Mode $= 8$ (8 students, the highest frequency). Total students $= 3 + 5 + 8 + 4 + 2 = 22$.

10 **[Intermediate]** • **Target: Reading tables**

Using the same quiz data, find the mean score to two decimal places.

Sum $= 5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170$. Mean $= \frac{170}{22} \approx 7.73$.

11 **[Intermediate]** • **Target: Reading bar charts**

The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.

Total $= 45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.

12 **[Intermediate]** • **Target: Reading bar charts**

Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

$\frac{30}{110} \times 100\% \approx 27\%$.

13 **[Intermediate]** • **Target: Reading bar charts**

The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

Mean $= \frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60$ mm.

14 **[Intermediate]** • **Target: Reading bar charts**

Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.

Decrease $= 80 - 55 = 25$ mm. $\frac{25}{80} \times 100\% = 31.25\%$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Word Problems & Reasoning

A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.

$P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Arithmetic & Number Sense

A fair six-sided die is rolled. Find $P(\text{a number greater than } 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.

$P = \frac{13}{52} = \frac{1}{4}$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Algebra Essentials

A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.

Even outcomes $\{2, 4, 6, 8\}$: $P = \frac{4}{8} = \frac{1}{2}$.

19 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 4.01 Geometry & Measurement

A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.

$P(\text{red}) = \frac{3}{10}$, so $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.

20 [Challenging] • Target: Combined and complementary events • Cross-Lesson: 5.01 Quantitative Comparisons

Two fair coins are tossed. Find $P(\text{both land heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Mean, median, mode and range

A data set of 6 values has mean 25. Five of the values are $20, 22, 28, 30, 32$. Find the sixth value.

a. 18 ✓

b. 28

c. 8

d. 150

Total needed $= 6 \times 25 = 150$; sum of five $= 132$; sixth $= 150 - 132 = 18$.

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22 [Foundational] • Target: MCQ — Mean, median, mode and range

The bar chart shows quiz scores. Find the total number of students, and the modal score.

a. 20 **students; mode** $= 8$ ✓

b. 20 students; mode $= 6$

c. 20 students; mode $= 9$

d. 18 students; mode $= 8$

Total $= 2 + 4 + 6 + 5 + 3 = 20$; the score 8 has the highest frequency (6 students).

23 [Foundational] • Target: MCQ — Reading tables

A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

a. $\frac{18}{60}$

b. $\frac{22}{60}$

c. $\frac{11}{30}$ ✓

d. $\frac{11}{60}$

$\frac{22}{60}$ simplifies (divide by 2) to $\frac{11}{30}$.

24 [Foundational] • Target: MCQ — Reading tables

Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?

a. 43% ✓

b. 30%

c. 50%

d. 26%

$(18 + 8) \div 60 = 26 \div 60 \approx 43\%$.

25 [Foundational] • Target: MCQ — Reading tables

A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.

a. 34

b. 30 ✓

c. 26

d. 25

$4 + 7 + 9 + 6 + 4 = 30$.

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26 [Foundational] • Target: MCQ — Reading tables

Using the same test data, find the mean score to two decimal places.

a. 79.67 ✓

b. 80.00

c. 76.33

d. 79.00

Sum $= 240 + 490 + 720 + 540 + 400 = 2{,}390$; mean $= 2{,}390 \div 30 \approx 79.67$.

27 [Foundational] • Target: MCQ — Reading tables

A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

a. 0

b. 1.2

c. 2

d. 1 ✓

With 25 values sorted, the median is the 13th value, which falls in the '1 pet' group.

28 [Foundational] • Target: MCQ — Reading tables

Using the same pet data, find the mean number of pets per family, to two decimal places.

a. 1.00

b. 1.20 ✓

c. 1.50

d. 30

Sum $= 0(6) + 1(10) + 2(7) + 3(2) = 30$; mean $= 30 \div 25 = 1.20$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Reading bar charts

The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.

a. 120; Romance

b. 100; Fiction

c. 120; Mystery

d. 120**; Fiction** ✓

Total $= 45 + 30 + 25 + 20 = 120$; Fiction had the most sales (45).

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30 [Intermediate] • Target: MCQ — Reading bar charts

Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.

a. 75%

b. 43% ✓

c. 30%

d. 40%

Decrease $= 70 - 40 = 30$; $30 \div 70 \approx 43\%$.

31 [Intermediate] • Target: MCQ — Reading bar charts

The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

a. 55; Store B

b. 55; **Store D** ✓

c. 140; Store D

d. 35; Store D

Range $= 140 - 85 = 55$; Store D had the highest sales (140).

32 [Intermediate] • Target: MCQ — Reading bar charts

The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.

a. 12; 2

b. 12; 2.4 ✓

c. 12; 3

d. 10; 2

Total $= 2 + 3 + 1 + 4 + 2 = 12$; mean $= 12 \div 5 = 2.4$.

33 [Intermediate] • Target: MCQ — Reading bar charts

The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.

a. 40; 10 ✓

b. 40; 7

c. 35; 10

d. 40; 15

Total $= 12 + 8 + 15 + 5 = 40$; range $= 15 - 5 = 10$.

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34 [Intermediate] • Target: MCQ — Basic probability

A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

a. $\frac{1}{5}$

b. $\frac{6}{9}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$ ✓

6 out of 15 simplifies to $\frac{2}{5}$.

35 [Intermediate] • Target: MCQ — Combined and complementary events

A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{1}{16}$ ✓

$\left(\frac{13}{52}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$.

36 [Intermediate] • Target: MCQ — Combined and complementary events

A bag holds 3 red and 4 blue marbles. Two are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{3}{7}$

b. $\frac{9}{49}$

c. $\frac{2}{7}$

d. $\frac{1}{7}$ ✓

$\frac{3}{7} \times \frac{2}{6} = \frac{6}{42} = \frac{1}{7}$.

37 [Intermediate] • Target: MCQ — Weighted averages

A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.

a. 84.50

b. 78.00

c. 91.00

d. 79.00 ✓

New total $= 12(78) + 91 = 1{,}027$; new average $= 1{,}027 \div 13 = 79.00$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Weighted averages • Cross-Lesson: 3.01 Algebra Essentials

Class A has 25 students with mean score 70; Class B has 15 students with mean score 90. Find the combined mean.

a. 80

b. 75

c. 160

d. 77.5 ✓

$\frac{25(70)+15(90)}{40} = \frac{3{,}100}{40} = 77.5.$

39 [Challenging] • Target: MCQ — Weighted averages • Cross-Lesson: 4.01 Geometry & Measurement

A student's final grade is 30% homework average (score 85) and 70% exam average (score 75). Find the final grade.

a. 82

b. 80

c. 78 ✓

d. 75.5

$0.3(85) + 0.7(75) = 25.5 + 52.5 = 78.$

40 [Challenging] • Target: MCQ — Weighted averages • Cross-Lesson: 5.01 Quantitative Comparisons

A survey combines two groups: 20 people with average age 30, and 30 people with average age 40. Find the combined average age.

a. 70

b. 35

c. 36 ✓

d. 34

$\frac{20(30)+30(40)}{50} = \frac{1{,}800}{50} = 36.$

41 [Challenging] • Target: MCQ — Weighted averages • Cross-Lesson: 6.01 Sequences & Patterns

A class of 8 students has an average score of 65. A 9th student joins, changing the new average to 68. Find the 9th student's score.

a. 71

b. 65

c. 88

d. 92 ✓

New total $= 9(68) = 612$; old total $= 8(65) = 520$; 9th score $= 612 - 520 = 92$.

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42 [Challenging] • Target: MCQ — Weighted averages • Cross-Lesson: 8.01 Word Problems & Reasoning

Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n.

a. 12

b. 15 ✓

c. 10

d. 20

$\frac{600+90n}{10+n} = 78 \Rightarrow 600 + 90n = 780 + 78n \Rightarrow 12n = 180 \Rightarrow n = 15$.

43 [Challenging] • Target: MCQ — Comparing data sets • Cross-Lesson: 1.01 Arithmetic & Number Sense

Team X scored $18, 22, 25, 23$ points in 4 games. Team Y scored $15, 15, 32, 26$. Both teams have the same mean. Which team is more consistent (smaller range)?

a. **Team X** ✓

b. Team Y

c. Cannot be determined

d. They are equally consistent

Both have mean 22. Range X $= 7$; range Y $= 17$. Team X is more consistent.

44 [Challenging] • Target: MCQ — Comparing data sets • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Class 1 has 16 students with mean score 70; Class 2 has 24 students with mean score 80. Find the combined mean of both classes.

a. 150

b. 75

c. 76 ✓

d. 77

$\frac{16(70) + 24(80)}{40} = \frac{3{,}040}{40} = 76$.

45 [Challenging] • Target: MCQ — Comparing data sets • Cross-Lesson: 3.01 Algebra Essentials

Data set A: $5, 10, 15, 20, 25$. Data set B: $13, 14, 15, 16, 17$. Both have mean 15. Which has the smaller range, and what is it?

a. Data set B; 20

b. **Data set B;** 4 ✓

c. Data set A; 4

d. Data set A; 20

Range A $= 20$; range B $= 4$. Data set B is smaller.

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